Week 2 build card

A cargo tray that keeps its load

Team ____ Builders A ____ B ____ C ____

Build a removable cart/tray with an edge that keeps the cube aboard. **It must not hinder the robot's movement.**

Materials: Beams and pins or folded pre-cut card, tape, one lightweight cube, and the mounting/clearance template.

Try it

1. Choose where the cube will sit and build a flat support.

2. Add an edge or rail without making the tray touch the floor or wheels.

3. Tilt the tray gently by hand and compare one change to its edge.

Our jobs

Builder connects pieces. Designer chooses/finds parts and checks the plan. Tester performs the hand test and reports one result. Everyone builds. Rotate when the mentor calls time.

Test and improve

Cube stays aboard during a gentle tilt. The cart fits securely, clears the wheels and floor, and cannot snag cables or block controls. A mentor checks a short run; if it rubs, drags or shifts, fix the build.

We noticed _____. We changed _____.

A mentor supplies an example tray for the first robot turn. Keep payload and tray unchanged during each distance comparison; test the team tray during the retest.

If time allows: after the core trials, attach the checked cart for the book-parking challenge. Help the programmers measure from its frontmost part and leave a gap before the book. The mentor chooses and supervises the tabletop route.

Fit attachments only to a stopped robot after the mentor checks clearance. Leave wheels, buttons, cables and sensors clear.